# YZV202E Optimization for Data Science
# Homework II

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## Homework Rules

- You must submit your solution as a PDF file and an .ipynb (Jupyter Notebook) solution file for programming questions.
- If you would like to, you can add hand-written solutions to the report instead of writing in LaTeX.
- Solutions must be self-contained and include the necessary intermediate steps to find the solution.
- Do not share any text that can be submitted as a part of an assignment (discussing ideas is okay).
- You may discuss the problems at an abstract level with your classmates, but you should not **share or copy** from your classmates or the Internet. You should submit your **own and individual** homework.
- Only electronic submissions that are sent before the deadline through Ninova will be accepted.
- You can use Desmos or Python for plotting.
- If you have any questions about the homework, you can send an email to Abdullah Kavaklı (kavakli23@itu.edu.tr) or Erhan Biçer (bicer21@itu.edu.tr).
- Reference books: Chong, E. K. P., & Zak, S. H. *An Introduction to Optimization* (2004); Kochenderfer, M. J., & Wheeler, T. A. *Algorithms for Optimization.*

## Question 1 :

You are given 2D data points from `data.txt`. Fit the best ellipse to these points by minimizing the residuals using a nonlinear least squares approach.

Use the rotated ellipse model given below:

$$\left(\frac{(x-h)\cos\theta + (y-k)\sin\theta}{\alpha}\right)^2 + \left(\frac{-(x-h)\sin\theta + (y-k)\cos\theta}{\beta}\right)^2 - 1 = 0$$

where $h, k$ are the center coordinates, $\alpha, \beta$ are the semi-axes, and $\theta$ is the rotation angle.

Estimate the parameters $h, k, \alpha, \beta, \theta$ using nonlinear least squares regression as using these steps:

(a) Derive the Jacobian matrix of the residuals with respect to all five parameters. Use this to linearize the problem at any given iteration (i.e., use first-order Taylor approximation).

(b) Implement and compare two Non-linear Least-Squares algorithms:

- Gauss–Newton method
- Levenberg–Marquardt method

For each, visualize the fitted curve and points and report:

- Number of iterations until convergence
- Final sum-of-squares loss value
- Condition number of the Jacobian at convergence

## Question 2 :

Consider the minimization of the following multimodal test function:

- Dolphin function:

$$f(x, y) = -\left|\sin(x-a)\cos(y-b)\right| - 4\exp\left(-\left(0.5(x-a)^2 + 0.5(y-b)^2\right)\right)$$

Validate the algorithms by solving the given function. Let $a$ be the 3rd from the last digit of your student number, and $b$ be the 2nd digit from the end. Use these values to customize the problem parameters accordingly.

1. Particle Swarm Optimization
2. Genetic Algorithm

Additionally, the following requirements must be fulfilled:

(a) Visualize the given function using 3D surface and 2D contour plots.

(b) Run each algorithm for 25 independent trials using different random seeds. Report the mean and standard deviation of the best fitness values, along with the success rate (defined as reaching a function value within a value (i.e $(10^{-3})$).

(c) For each trial, randomly sample a rectangular subregion of the function's default domain (e.g., within the square domain defined as $x, y \in [-10, 10]$). Justify and discuss the impact of this constraint on algorithmic performance.

(d) Add Gaussian noise $\mathcal{N}(0, \sigma^2)$ to the function evaluations $f(x, y)$, where $\sigma = 0.01$ X (range of the function). Re-run the algorithms under these noisy conditions and analyze the robustness of each method.

(e) Limit each algorithm to a maximum of 750, 1000, and 1500 function evaluations per run. Report the algorithm's performance under each of these budget constraints, and discuss any trade-offs observed.

Finally, briefly address the following questions:

(f) Which algorithm was more successful under stochastic and noisy conditions?

(g) Which function posed greater difficulty for the optimization process, and why?

Your answers must be supported by both theoretical reasoning and explicit references to your experimental findings. Purely speculative or unsupported answers will not be accepted.

## Question 3 :

Consider the following non-convex function (Beale's Function):

$$f(x_1, x_2) = (1.5 - x_1 + x_1 x_2)^2 + (2.25 - x_1 + x_1 x_2^2)^2 + (2.625 - x_1 + x_1 x_2^3)^2$$

Global Minimum = (3, 0.5)

(a) Implement both the Fletcher–Reeves Conjugate Gradient and Newton's methods to minimize the above function. Visualize their convergence trajectories on the function's contour plot, starting from at least three distinct initial points: one close to the global minimum, one far from it, and one that results in slow convergence or failure. How can such a challenging initial point be identified? Briefly discuss your reasoning and support your hypothesis with **additional experiments.**

(b) Analyze the local curvature by computing and visualizing the Hessian condition number $\kappa(H_f(x))$ at various points along each path (Chosen points in (a)). Discuss how it affects Newton's performance.

(c) Which method would you recommend for optimizing non-convex functions like this one? Your answer must be supported by both theoretical reasoning and explicit references to your experimental findings. Purely speculative or unsupported answers will not be accepted.